



East West University

Department of Computer Science
and Engineering

Project Proposal

Course: Software Engineering

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Section: 05

Submitted by:

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Project Title:An AI Job-Hunting Agent

Project overview:The system is designed to assist job seekers by automatically finding and recommending suitable job opportunities using an AI-powered agent. As a job seeker, the user can upload a CV which the system analyzes to extract skills, experience level, job roles and location preferences in order to create a personalized profile. Based on this profile, the AI agent navigates predefined job portals such as Bdjobs and LinkedIn-style platforms by simulating human interactions like searching, scrolling, pagination and infinite scrolling. The agent observes page structures and adapts its search behavior using AI reasoning rather than relying only on fixed rules. The system extracts structured job information including job title, company, location, requirements and descriptions while handling dynamically loaded content and filtering out duplicate postings. The extracted jobs are compared with the user's profile to generate match percentages and suitability scores allowing jobs to be ranked by relevance. The system also identifies suspicious or potentially fake job postings using a combination of LLM-based reasoning and factual validation. After completing the search, the AI provides a summarized recommendation explaining which jobs best match the user's profile and why along with application deadline reminders to support timely decision-making.

Purpose of the Project:The purpose of this project is to design and develop an AI-powered job search agent that helps job seekers find suitable job opportunities efficiently and intelligently. The system aims to reduce the time and effort required to manually search multiple job portals by automating job discovery, analysis, and recommendation. By understanding a user's CV and preferences through a Large Language Model (LLM), the system provides personalized job matching, ranks opportunities based on suitability, and explains recommendations in a clear and understandable manner. The project also aims to demonstrate the practical application of AI agents, LLM reasoning, and web automation to solve a real-world problem in the job search domain.

Scope of the Project:

In Scope: The scope of this project includes CV upload and AI-based profile generation, automated navigation of predefined job portals such as Bdjobs and LinkedIn-style platforms, and extraction of structured job information from dynamically loaded web pages. The system will match and rank job listings based on user skills, experience, and preferences, filter duplicate postings, and detect suspicious or potentially fake jobs using AI reasoning and factual checks. Additional features within scope include pagination and infinite-scroll handling, job application deadline reminders, AI-generated job recommendations, and adaptive learning through site pattern memory.

Out Scope:

The project scope is limited to a selected number of supported job websites and does not guarantee full automation of job applications. Advanced features such as automatic application submission, real-time employer communication, and unrestricted access to all job portals are considered future enhancements and are not included in the current scope.

Stakeholders: Stakeholders are people who are related to or affected by the system.

1. Primary stakeholders:

- **Job Seekers:** Job seekers are the primary users of the system. They interact directly with the platform by uploading CVs, defining job preferences, reviewing job recommendations, and receiving application reminders. The system is primarily designed to meet their needs by automating job discovery, matching, and recommendation.
- **Admin:** The system administrator is responsible for configuring supported job portals, monitoring system operations, handling failures or blocked searches, and ensuring data accuracy and system availability. The administrator plays a critical role in maintaining system reliability and performance.

2.Secondary Stakeholders:

- **Job Portal:**Job portal service providers (e.g., Bdjobs, LinkedIn-style platforms) act as external data sources for the system. While they do not directly interact with the system, their website structures, content policies, and availability affect data access and system behavior.
- **Academic Supervisor:**Academic supervisors and evaluators oversee the project's progress and assess its technical quality, documentation, and learning outcomes. Their feedback influences system design decisions and project evaluation.
- **Employers:**Employers are organizations or recruiters who publish job postings on supported job portals. Although they do not directly interact with the system, they are affected by it because their job listings are analyzed, matched, and recommended to suitable candidates, potentially improving the quality of applicants they receive.

Stakeholder Needs & Analysis:

The stakeholders were asked several questions to understand their needs and expectations from the system. Job seekers were asked about their education level, field of interest, skills, years of experience, preferred job type (full-time, part-time, remote), desired location, expected salary, willingness to relocate, preferred company type, and whether they wanted notifications for new jobs. From this analysis, it is clear that the system must provide accurate job matching, personalized recommendations, and user-friendly interactions. The AI agent should be capable of analyzing CVs, comparing skills with job requirements, ranking opportunities, and summarizing recommendations in a way that supports efficient decision-making. This ensures that job seekers can maximize their chances of finding suitable positions while minimizing the effort and time spent on manual job searches.

List of Requirements:

1. Functional requirement:

- CV Understanding & Profiling
- Supported job site Navigation
- Ai-guided search execution
- Job data extraction
- Deadline reminder
- Fake job detection
- Output-based job recommendation
- Duplicate job filtering
- Site pattern memory

2. Non-Functional Requirement:

- Process CV and extract info within 30 seconds
- Complete 3-portal search within 5 minutes
- Support 100+ concurrent users
- 99.5% uptime availability
- Handle growth from 100 to 10,000 users

Extraordinary requirements (wow factors): These features make the system more attractive.

- Skill gap analyzer
- Job market intelligence dashboard
- Adaptive search strategy

House of Quality(QFD Integration):

Customer Requirements:

- Save time in job searching
- Find relevant job matches
- Avoid fake jobs
- Never miss deadlines
- Understand skill gaps
- No duplicate jobs

- Easy to use
- Secure CV handling
- Personalized recommendations
- Market trends insight
- Reliable system
- Multi-site search

Technical Requirements:

- CV Parsing Accuracy
- Search Completion Time
- Job Matching Relevance
- Fraud Detection Precision
- Duplicate Detection Rate
- System Uptime
- Response Time
- Data Security
- UI Usability Score
- Scalability
- API Reliability
- Browser Automation Success

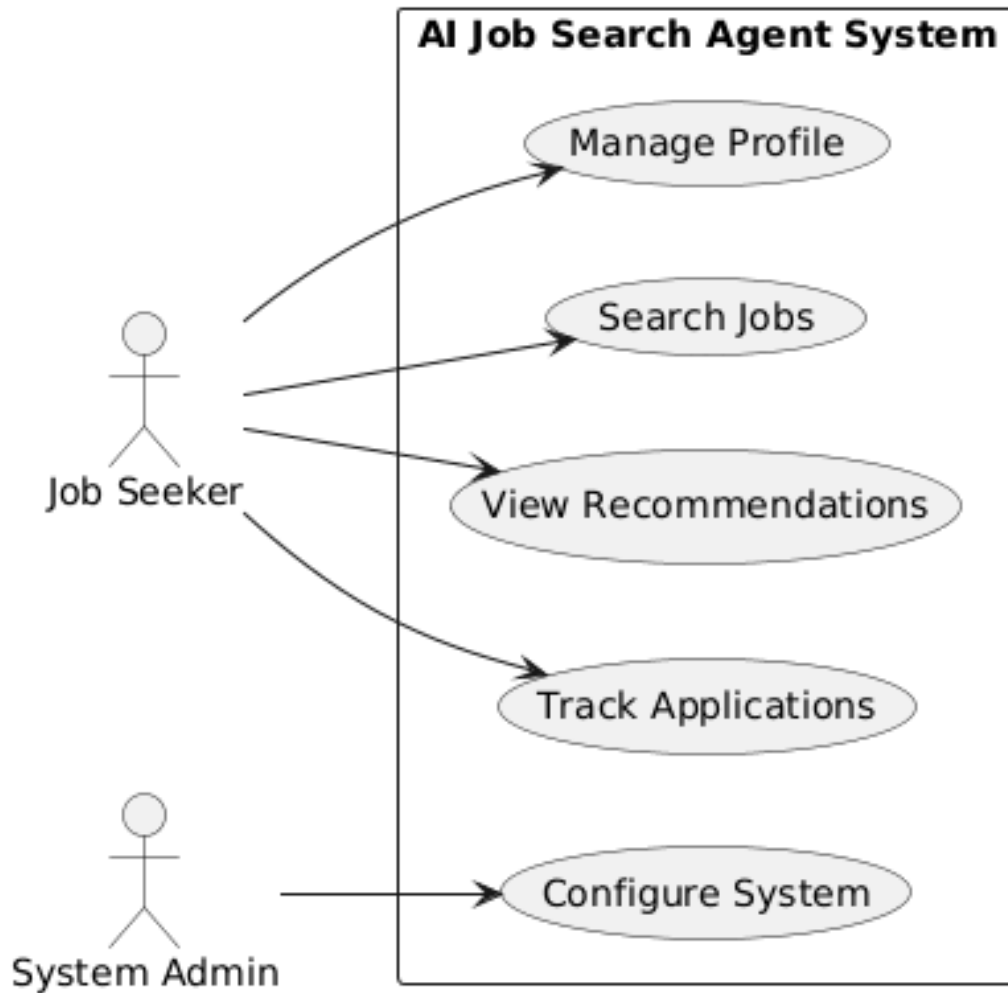
QFD Matrix (House of Quality):

Customer Requirements (CRs)	Technical Requirements (TRs)	Relationship (Strong /Medium/Weak)
CR1:Save time in job searching	TR1: CV Parsing Accuracy	Strong
CR2:Find relevant job matches	TR2:Search Completion Time	Strong
CR3:Avoid fake jobs	TR3: Job Matching Relevance	Strong
CR4:Never miss deadlines	TR4: Fraud Detection Precision	Strong
CR5:Understand skill gaps	TR5: Duplicate Detection Rate	Medium
CR6: No duplicate jobs	TR6:System Uptime	Medium
CR7:Easy to use	TR7:Response Time	Medium
CR8:Secure CV handling	TR8:Data Security	Strong
CR9:Personalized recommendations	TR9:UI Usability Score	Medium

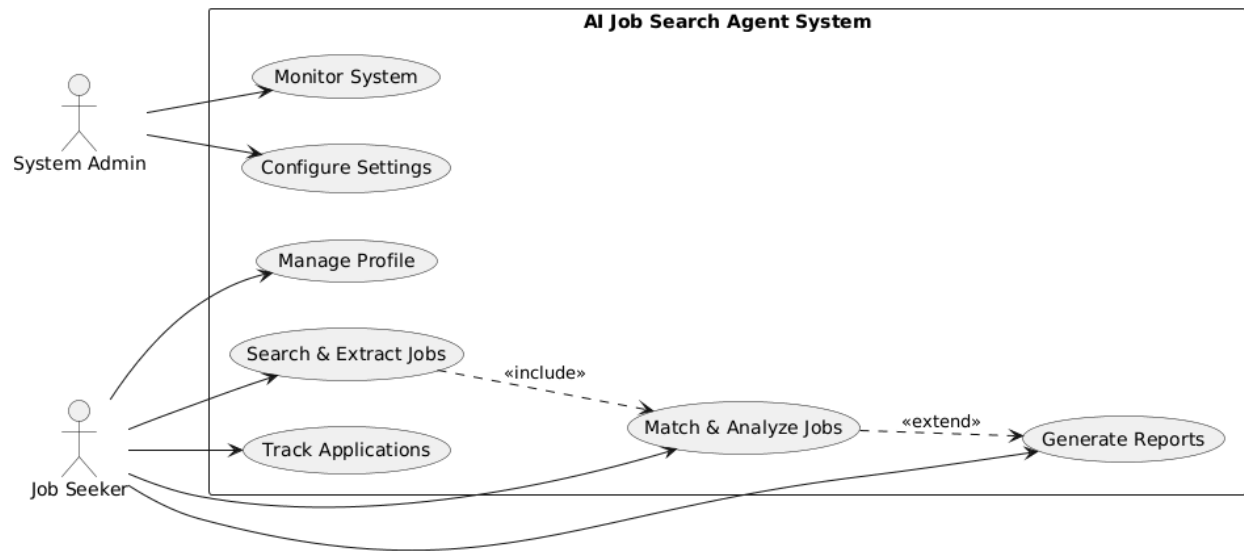
CR10:Market trends insight	TR10:Scalability	Medium
CR11:Reliable system	TR11:API Reliability	Medium
CR12:Multi-site search	TR12:Browser Automation Success	Medium

Use case diagram:

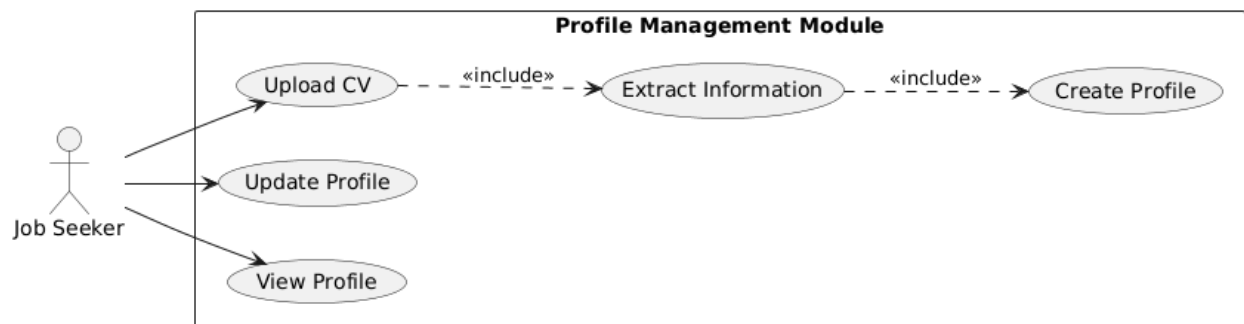
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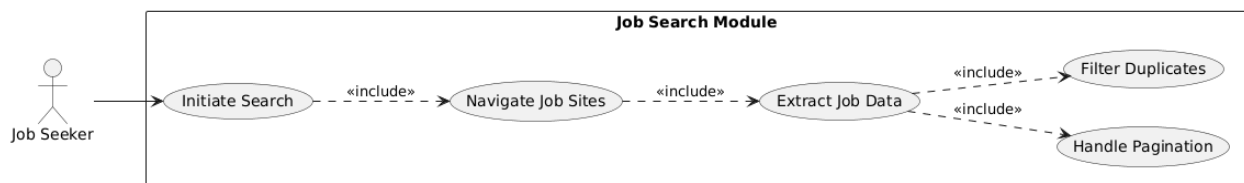
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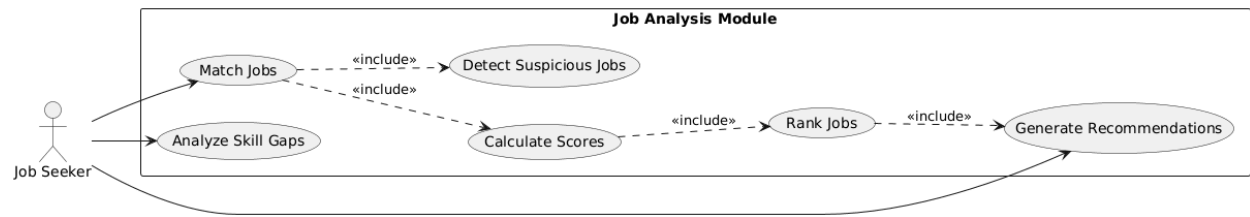
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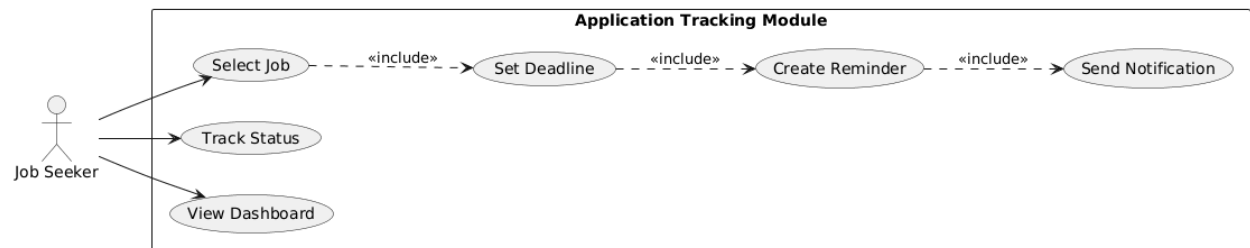
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Level 1.3:

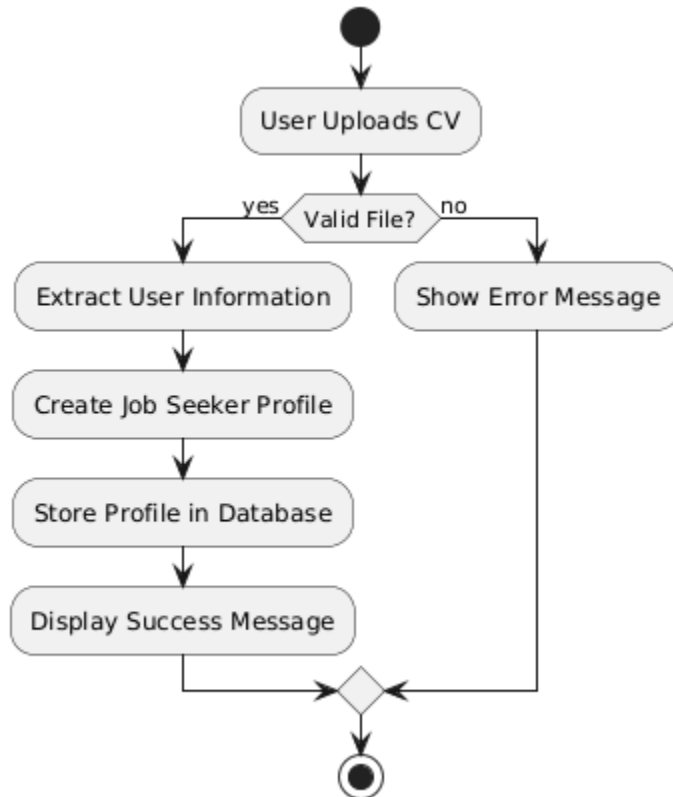


Level 1.4:

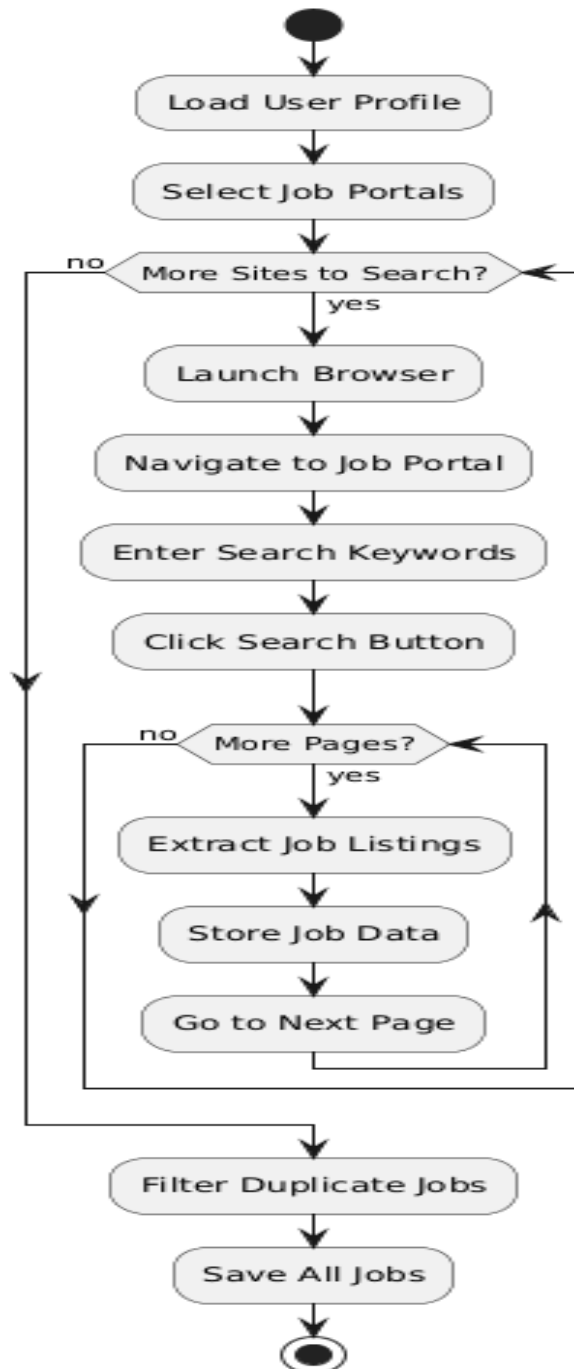


Activity Diagram:

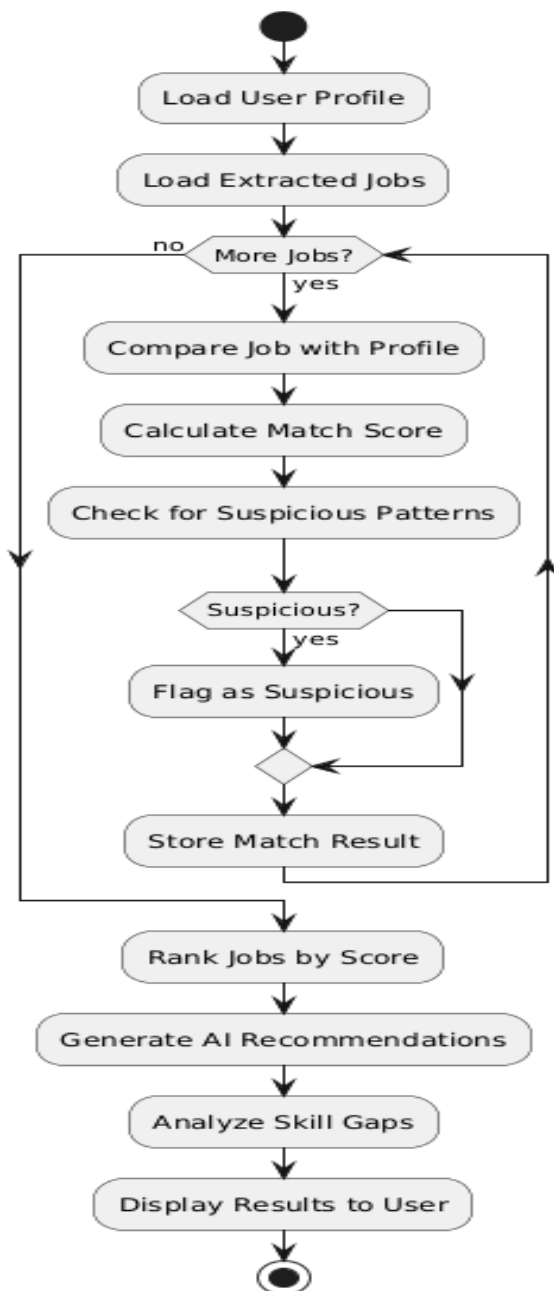
User Profile creation:



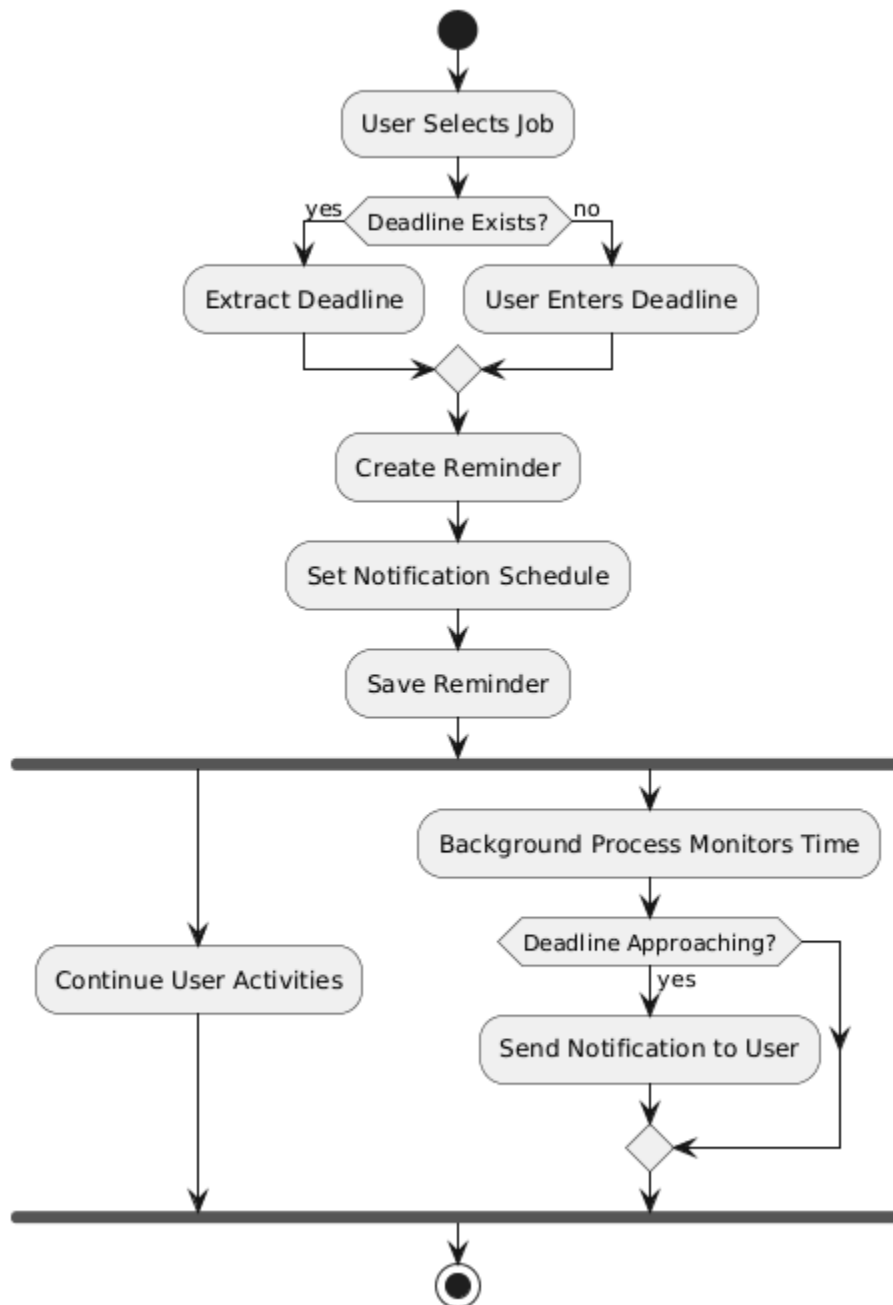
Job search & extraction:



Job matching & recommendation:

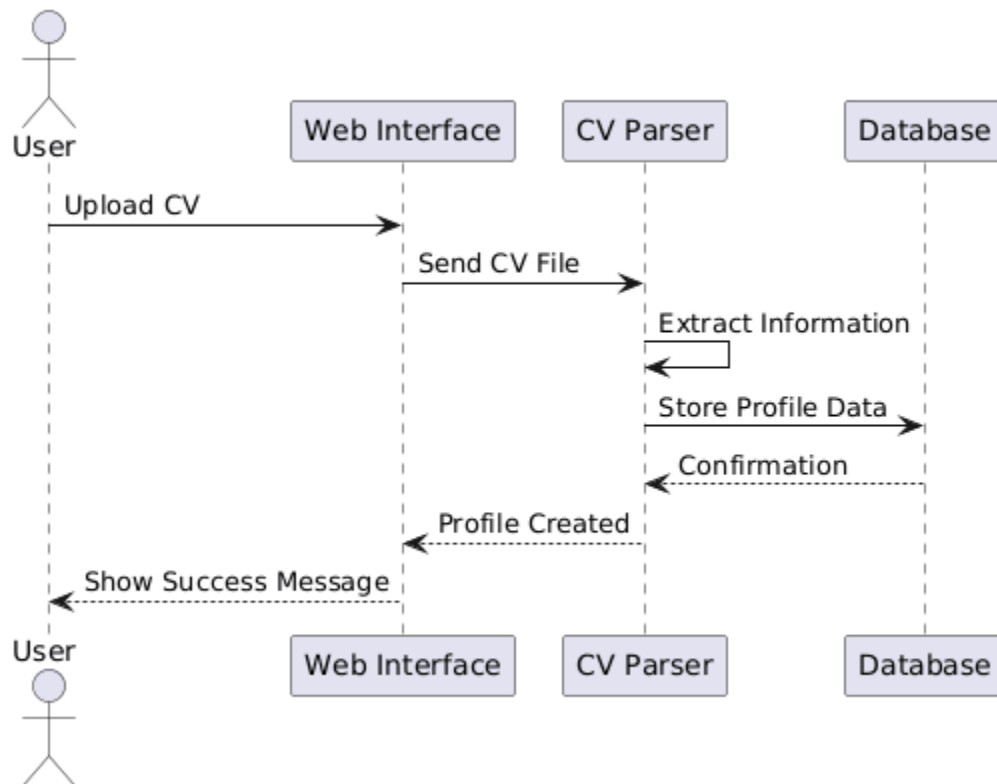


Application tracking & reminders:

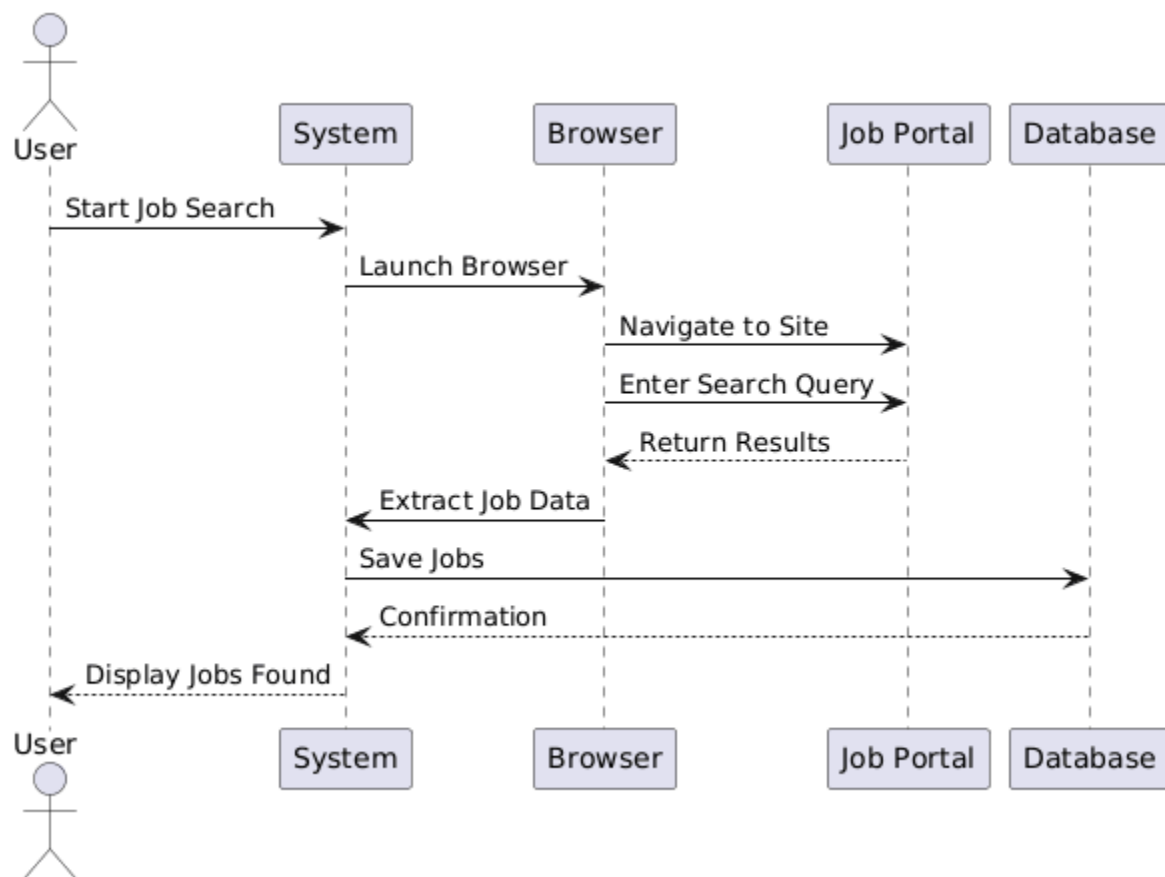


Sequence Diagram:

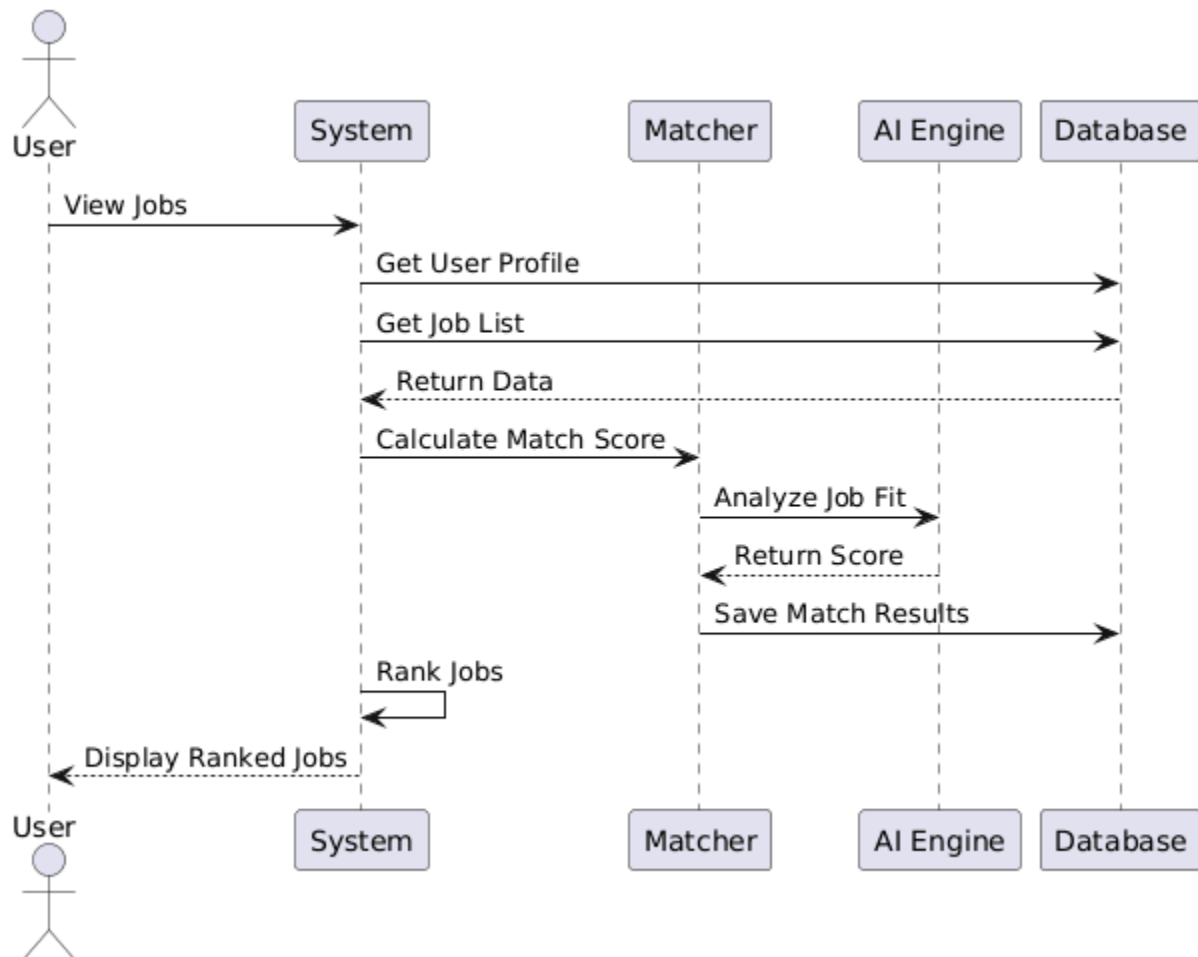
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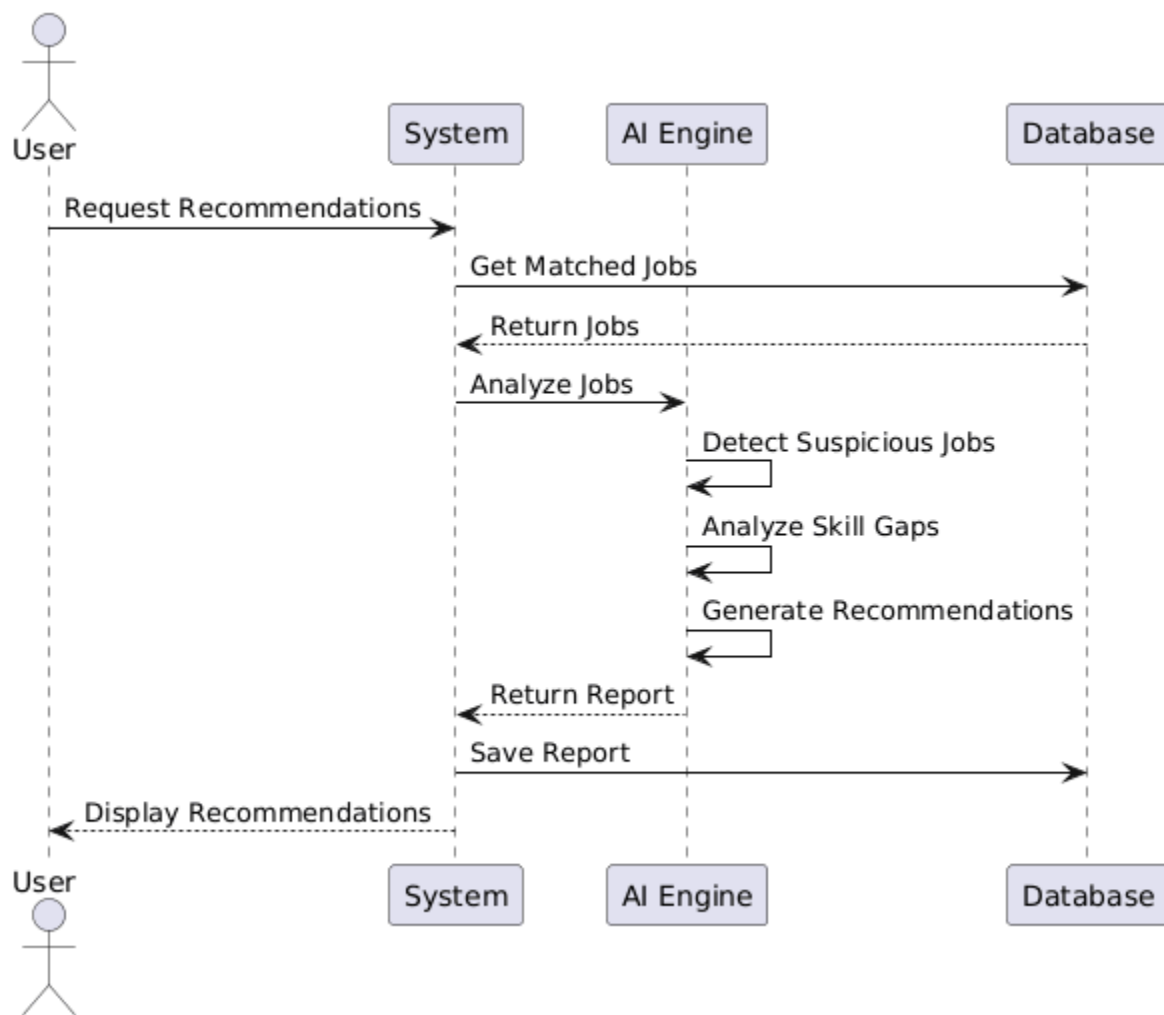
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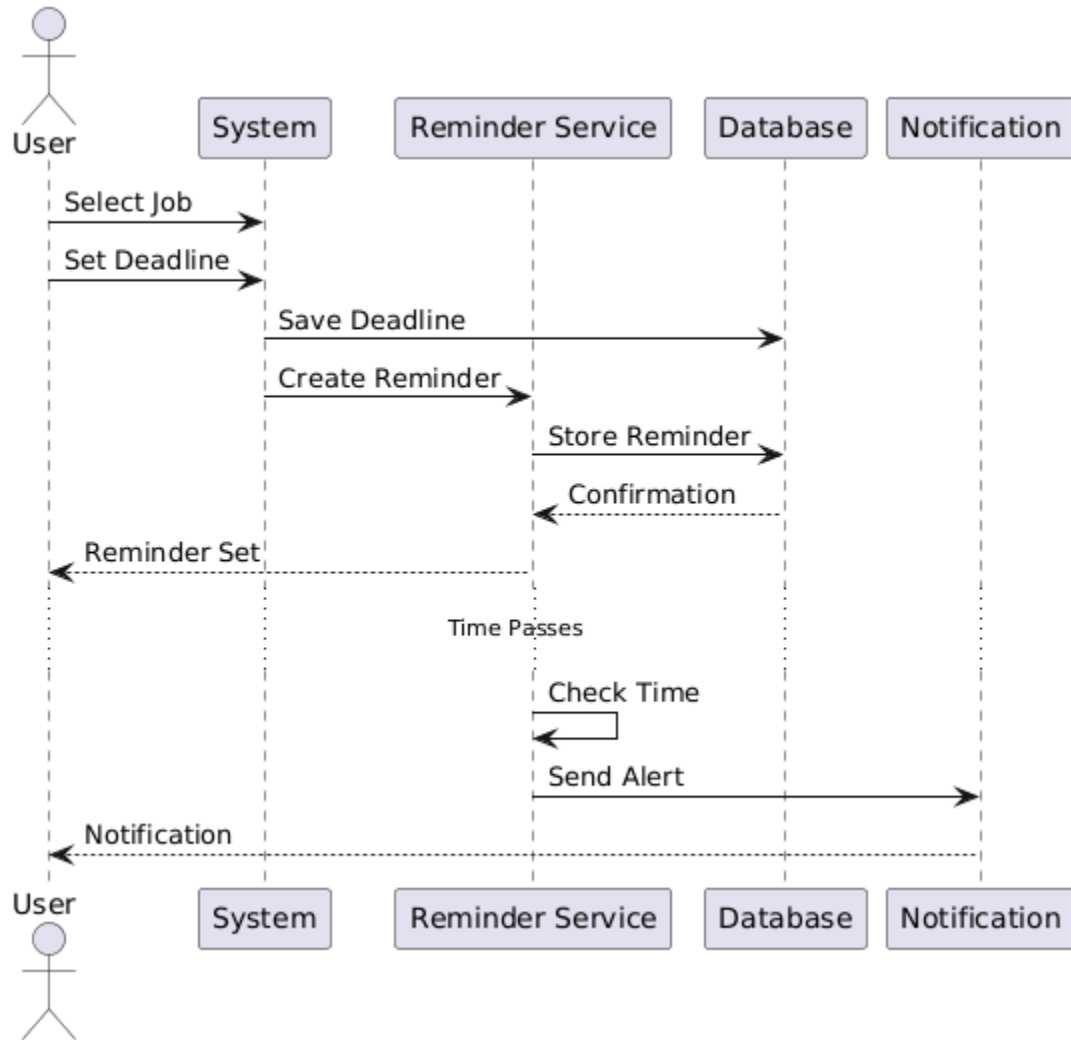
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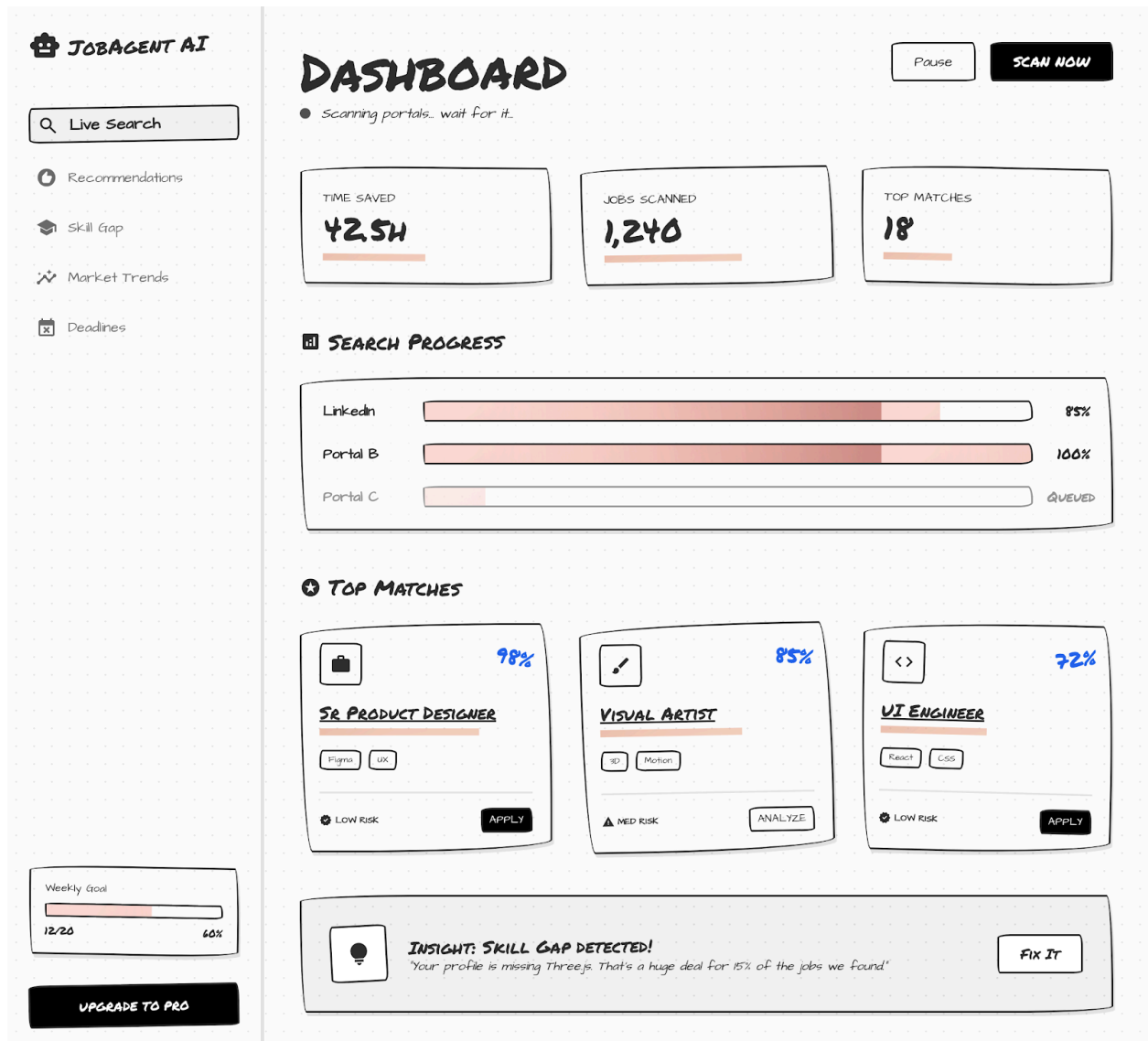


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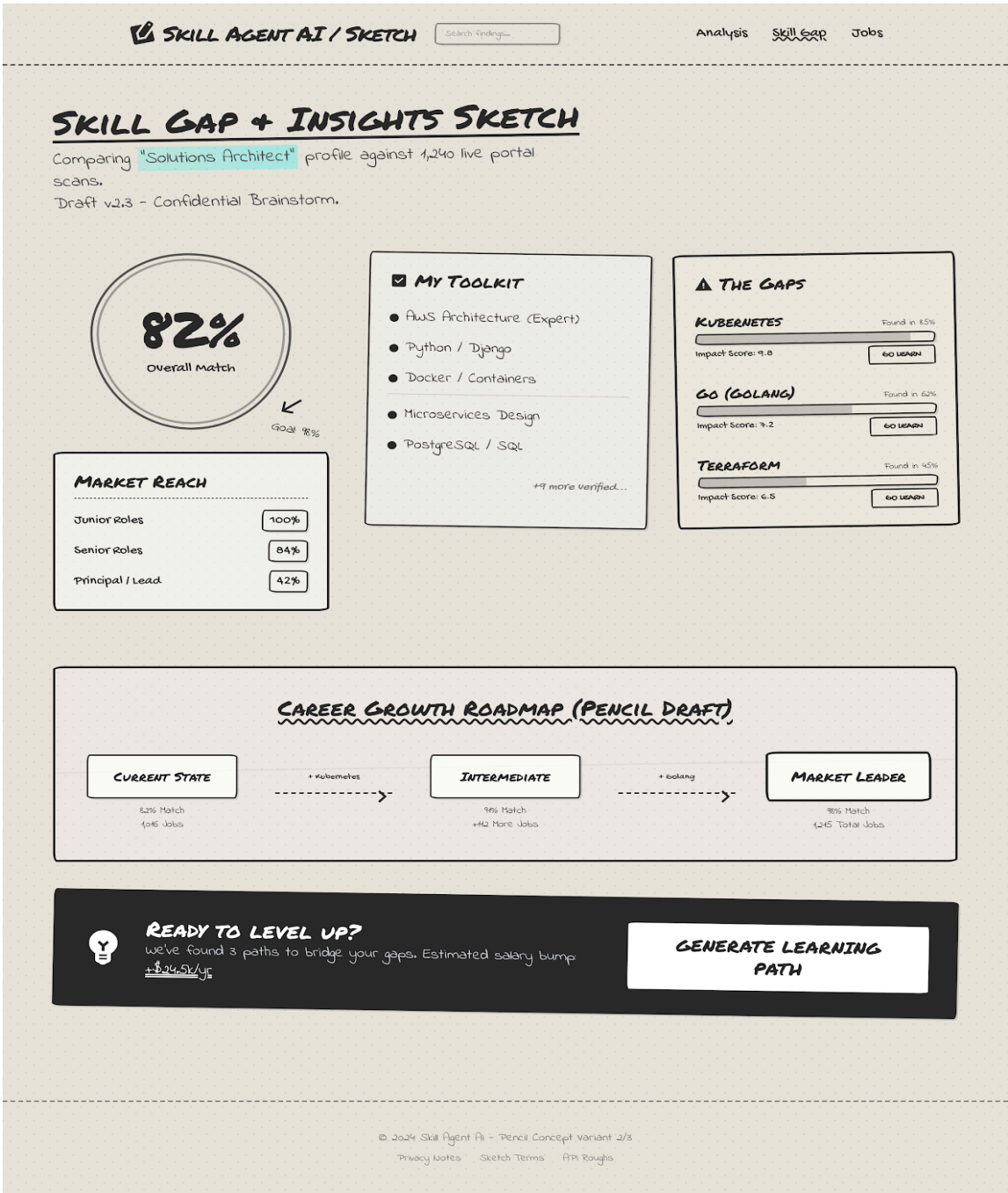


UI sketches:

1.



2.



3.

